

Waibhav Jha

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Profile

Adept in Machine Learning, Deep Learning, and Full Stack Development. Proficient in Python, Java, Big Data, and Web Technologies with hands-on experience building ML models, data pipelines, and scalable applications. Skilled in data analysis, model development, and collaborative research-driven problem solving.

Education

Amrita Vishwa Vidyapeetham, Coimbatore
B.Tech in Computer Science and Artificial Intelligence

2023 – Present
CGPA: 7.87/10

Skills

- **Programming:** Python, Java, SQL, HTML, CSS, JavaScript, Apache Spark
- **Domains:** Machine Learning, Deep Learning, Computer Vision, Big Data Analytics, Time Series Analysis, Computational Biology
- **Tools:** TensorFlow, PyTorch, Scikit-learn, OpenCV, Git, GitHub
- **Web Development:** REST APIs, Frontend Development, Backend Integration
- **Soft Skills:** Analytical Thinking, Team Collaboration, Communication, Problem Solving

Selected Projects

- **Physics-Based Vision Language Model for CFD Inference** Developed a multimodal transformer-based framework to extract physical parameters directly from computational fluid dynamics (CFD) simulation data. Implemented vision transformers to encode simulation frames and aligned visual embeddings with language models to generate captions describing fluid flow patterns.
- **Vision-Based Motion Prediction and Collision Risk Analysis for Autonomous Driving** Built a physics-aware computer vision pipeline using YOLO object detection, SORT tracking, and MiDaS depth estimation to model object trajectories in dynamic driving scenes. Implemented Monte Carlo time-to-collision simulations to estimate collision risk and generate interpretable trajectory and risk visualizations.
- **Blockchain Log Analytics for Detecting Fraudulent Transactions** Designed a distributed big data analytics pipeline using Hadoop HDFS and Apache Spark to analyze Ethereum transaction logs. Extracted graph-based wallet interaction features and applied Spark MLlib clustering to identify anomalous wallets exhibiting suspicious transaction behavior.
- **Early Warning Signals and Hybrid Time Series–Deep Learning Models for Climate Tipping Point Prediction** Developed a research-grade forecasting framework combining classical time series models (ARIMA, SARIMA, GARCH) with deep learning models (LSTM, GRU, CNN-LSTM). Designed a Climate Instability Index using rolling variance and autocorrelation to detect early warning signals of potential climate tipping points.
- **Personalized Pet-Grooming Recommendation Website** Developed a survey-based recommendation platform using HTML, CSS, and JavaScript that suggests personalized grooming solutions based on pet characteristics and owner preferences.
- **Portfolio Optimization using Monte Carlo Simulation and LSTM** Built a financial modeling system to analyze portfolio risk-return tradeoffs using Monte Carlo simulations and sequence modeling with LSTM networks. The research work was published at IEEE RAICS 2024.

Achievements

- **IEEE Publication:** “Portfolio Optimization using Monte Carlo Simulation and LSTM” – IEEE RAICS 2024.
- **Hackathon Achievement:** Top 100 Finalist in the International IBM TechXchange Hackathon.